

Lab 3: Architecture Justification

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Architecture Selection: We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network System.

Architectural Choice

Layered Architecture with three layers: Presentation (User Interface), Business (Session Manager, TimeCredit Service, Calendar Sync) and Data (Database). Each layer talks to the next only through interfaces: ISession, ICredit, ICalendar and IData.

Reasons for using this architecture:

1. The features are split cleanly into layers where browsing and booking screens belong to the UI, booking and credit rules to the business layer, and member, session and credit records to the database. This is easy to build and test for a platform of this size, and microservices would be too complex for it.
2. Credit rules stay in one place. Rules like “no booking with 0 credits” and “credits move only after both sides sign off” live in the business layer, so a web or mobile UI follows the same rules. The calendar or database part can be changed without touching the UI.

Security Advantage

The UI cannot reach the database directly. Every request goes through the Session Manager, which checks that the member is logged in and only allows them to change their own sessions. Credit balances can only be changed by the Time-Credit Service, so they cannot be edited from the UI.

Performance Benefit

All layers run inside one application, so calls between components have no network delay between services. Frequent actions like skill search and balance checks go straight to one database and respond quickly.